Achouri Youcef Islem

Applied AI and Machine Learning Internship

Sidi Bel Abbès, Algeria · yi.achouri@esi-sba.dz
youcef-achouri.vercel.app · github.com/youcef-ach · LinkedIn

Final-year ESI-SBA engineering student building model-driven software for language and spatial data. Available January 2027 for a final-year internship; duration flexible, generally three months or longer.

EDUCATION

ESI-SBA 2021 – 2027 (expected)
École Nationale Supérieure d'Informatique de Sidi Bel Abbès
State Engineering Degree in Computer Science (Bac+5) · Artificial Intelligence and Data Structures

PROFESSIONAL EXPERIENCE

Partner and Lead Spatial Systems Engineer March 2026 – Present
Azurid · AzuridTwinUp case study

- Built AzuridTwinUp, an authenticated industrial digital twin platform, with a Three.js/React viewer and NestJS/Prisma backend. Recorded viewer demonstration available.
- Worked across Matterport Pro3 LiDAR and DJI Mini 3 image acquisition, alignment troubleshooting in Metashape/RealityScan, E57 processing, and Meshopt/KTX2 delivery.

AI/GIS Project Lead February – May 2026
Institutional project placement · Algeria

- Led and supervised the team developing an AI/GIS application in collaboration with an institutional partner.
- Built the Arabic extraction module with Qwen 2.5 7B/Ollama, converting descriptions into nested JSON with syntax and schema validation.

Telecommunications Intern September 2025
Algérie Télécom · Batna, Algeria

- Studied fibre-optic and 5G network principles and explored potential AI applications for telecommunications development.

SELECTED PROJECTS

Enterprise Operations Intelligence Hub 2026
Python · LangGraph · FastAPI

- Owned agent orchestration in a four-person team: routed incidents to parallel department workflows and reconciled their decisions into an executive brief.
- Built deterministic mock execution for inspecting the same graph without API calls; enterprise actions are simulated.

Virtual staging on 3D scans 2026
Python · Mask2Former · SAM · Babylon.js

- Processed E57 scan geometry and poses; generated semantic labels across 21 real-scan viewpoints and inspected SAM boundary refinements on one scan face.
- Built a Babylon.js placement and occlusion viewer. The public interactive demo uses synthetic geometry; real scans provide separate model evidence.

TECHNICAL SKILLS

AI and data: Python, PyTorch, NumPy, LangGraph, Qwen, Ollama, Mask2Former, SAM.
Engineering: FastAPI, Django, React, TypeScript, NestJS, PostgreSQL, Docker, Git, pytest.

LEADERSHIP AND AWARDS

Vice President, INGENIUMS Club 2023 – 2024

- Co-organized technical and cultural events; invited artists and regional student delegations, coordinated teams, and presented projects to visitors and school leadership.

Awards: 2nd place, DevFest Hackathon 2023 · 3rd place, NCS Hack 3.0, June 2026.
Languages: Arabic (native), English (fluent), French (fluent).